

Kyumin Han

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EDUCATION

University of Pittsburgh, B.S. in Computer Science, Minor in Business

Expected Graduation: May 2028

GPA: 3.31 / 4.0

Pittsburgh, PA

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Systems Software, Web Applications

Member: Computer Science Club, AWS Cloud Club, Taekwondo Club

TECHNICAL SKILLS

Programming Languages: Java, Python, C, HTML, CSS, C++

Frameworks & Tools: Pandas, React.js, Vite.js, Node.js, Git, GitHub, Jupyter, AWS, Pandas, Anaconda, HuggingFace, OpenRouter

WORK EXPERIENCES

University of Pittsburgh

Pittsburgh, PA

Undergraduate Teaching Assistant

July 2026 - Present

- Host two open class labs for students to learn difficult topics about **Python** programming, topics, and ideas
- Held **6** hours of weekly office hours, mentoring students in Python to boost lab completion rates by **30%**
- Demonstrating leadership by teaching many students about various topics in programming, data structures, and algorithms.

PROJECTS

NVIDIA Verification Layer - SteelHacks XIII | Python, Jupyter Notebook, Nemotron Ultra and Super

Sept 2026

- Built a safety gate that audits LLM-generated medical advice. An LLM extracts findings, then a deterministic **Python** policy layer converts findings into a show/withhold decision, so every withheld answer names the rule that blocked it.
- Evaluated on a **41-case** labeled set built by injecting known flaws into correct drafts, making ground truth free. The gate caught **26/31** unsafe drafts while shipping **10/10** safe ones. The model's own risk score alone caught only **17/31** at identical coverage.
- A same-prompt GPT-4o-mini judge caught **30/31** but shipped only **2/10** safe answers - so recall alone can't distinguish a usable gate from one that blocks the product.

Multi-Lingual LLM Medical Research | Python, Qwen3-8B, Hugging Face, Jupyter Notebook

March 2025 – July 2025

- **Developed** a model-agnostic uncertainty estimation pipeline in **Python** (with Google Colab) using **Qwen3-8B** to classify ~5,000 medical triage scenarios as non-emergent, partially emergent, or emergent across English, Hindi, and Romanized Hindi inputs.
- **Engineered** a **2S** (semantic similarity) **confidence score** to quantify model certainty on critical emergency predictions across datasets. Wanted to clarify if the model was confident in its answers before proceeding.
- **Evaluated** cross-lingual consistency using **BERTScore** precision and recall metrics across identical prompts in multiple languages. Knowing how precise the model is allows me to factor out language barriers.
- **Found** systemic model overconfidence, identifying the LLM produced a **>75% confidence score** on nearly all 5,000 cases, regardless of input language.